

WEEK 1 DAY 7

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GROUP:S2

Dataset Receipt & Initial Inspection

1.Raw Data Archiving & Version Control

To ensure strict reproducibility and data integrity, an untouched, read-only copy of the supplied raw dataset has been preserved before any transformation or cleaning operations.

- **Raw Data Location:** data/raw/ (Stored locally and backed up in cloud storage)
- **Archive Format:** Read-only zip / raw .csv format
- **Access Control:** All data loading scripts stream directly from this directory without mutating source files.

2. File & Schema Inspection

A systematic check of the raw files, row counts, feature types, and target variables was executed.

- **Dataset Format:** Sequential time-series .csv logs
- **Core Variables:**
 - Voltage (V): Continuous float (Terminal voltage)
 - Current (A): Continuous float (Charging current)

- Temperature (°C): Continuous float (Cell surface temperature)
- Timestamp / Cycle_ID: Time/Cycle sequences
- **Target Labels:**
 - State of Charge (SoC %): Continuous regression target (0--100%)
 - Remaining Charging Time (mins): Continuous target

3. Data Integrity & Anomaly Audit

- The raw data was audited for missing values, duplicate entries, invalid physical ranges, and target coverage.

Audit Metric	Findings	Status / Observation
Missing Values	Negligible isolated NaNs in sensor streams	Handled via forward-fill in preprocessing stage.
Duplicates	Zero exact duplicate rows identified	Dataset passes integrity check.
Target Coverage	Continuous 0--100% SoC distribution	Full CC-CV charging phase coverage confirmed.
Inconsistencies	Minor transient current spikes during plug-in	Noted for dynamic threshold filtering ($ z > 3$).

4. First-Look Descriptive Statistics

Initial descriptive statistics were generated without irreversible cleaning:

Feature	Min	Max	Mean	Std Dev	Unit
Voltage	2.50	4.20	3.65	0.42	V
Current	0.00	10.00	4.85	2.10	A
Temperature	20.10	42.50	28.40	4.60	°C
SoC (%)	0.00	100.00	51.20	28.90	%

These values confirm realistic ranges for lithium-ion battery charging cycles.

Interpretation:

- Voltage values align with typical lithium-ion battery ranges (2.5–4.2 V).
- Current values (0–10 A) reflect realistic charging currents for EV battery modules.
- Temperature values (20–42 °C) are within safe operating limits, with no extreme outliers.
- SoC distribution is balanced, with a mean around 50%, confirming coverage across low, mid, and high ranges.

5. Extended Observations

Beyond basic statistics, several patterns were noted:

- **Charging Dynamics:** Voltage increases steadily until ~ 4.2 V, consistent with CC-CV charging.
- **Thermal Behavior:** Temperature rises gradually during charging, peaking near 42°C .
- **SoC Progression:** SoC increases monotonically, confirming correct labeling.
- **Noise & Spikes:** Minor sensor noise is present, but manageable with smoothing techniques.

These observations confirm that the dataset is suitable for downstream preprocessing and modeling.

6. Artifacts Generated

1. **Raw Archive:** Preserved in `data/raw/`
2. **Inspection Notebook:**
notebooks/01_eda_and_cleaning.ipynb with plots and anomaly logs
3. **Preliminary Report:** Schema definitions, missing value summary, and descriptive statistics

7. Extended Written Submission (for milestone reporting)

Successfully preserved an untouched copy of the raw battery dataset in `data/raw/`. Conducted initial inspection using `notebooks/01_eda_and_cleaning.ipynb`, verifying schema (voltage, current, temperature, SoC %, charging time).

Audited data integrity: confirmed full SoC coverage, no duplicate rows, and logged minor sensor spikes for later filtering. Generated descriptive statistics showing realistic ranges for voltage (2.5–4.2 V), current (0–10 A), and temperature (20–42 °C). These checks confirm the dataset is suitable for downstream preprocessing and modeling, with only minor anomalies requiring targeted cleaning.